

# RISHI KIRAN KARNATAKAM

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## Summary

Motivated Computer Science student seeking job and internship opportunities to apply my skills in software development and problem-solving. Quick to learn new technologies and adapt to challenges, focusing on contributing to innovative and impactful projects.

## Education

## Skills

- **Programming Languages:** C, Python, Dart, C#
- **Databases:** MySQL, MongoDB
- **Cloud Platforms:** Amazon Web Services (AWS)
- **Machine Learning/AI:** TensorFlow, TensorFlow Lite
- **Version Control:** Git, GitHub
- **Frameworks/Libraries:** Flask, Tkinter, Chess Library, Unity Engine, Flutter
- **DevOps Tools:** Jenkins

## Projects

- Built a mobile app for real-time plant disease detection using camera or gallery images.
- Integrated a TensorFlow Lite model (Inception V3) for accurate disease prediction with offline support.
- Enabled local data storage and automatic syncing to MongoDB Atlas when connected online.
- Designed a clean, modern UI with a sidebar to view history, profile, and AI predictions.
- Developed an AI-driven chess engine utilizing minimax, alpha-beta pruning, and Genetic Algorithms for optimized decision-making.
- Enhanced strategic depth and move accuracy by 20%.
- Achieved a 30% improvement in move generation speed through alpha-beta pruning.

- Developed a machine learning model to classify various diseases on cotton leaves through image processing techniques.
- Improved disease detection accuracy by 25% using transfer learning with pre-trained CNN models.
- Achieved 90%+ accuracy on test datasets, optimizing performance for agricultural disease diagnosis.

- Developed a 3D interactive solar system model to enhance gamified learning.
- Created a first-person 3D game inspired by Flappy Bird, leveraging dynamic renderings and immersive gameplay elements.

## Awards

- **National Hackathon on Generative AI** – Achieved 2nd place in a National Level Hackathon focused on Generative AI. (Sep 2024)
- **Internal Hackathon on Generative AI** – Achieved 3rd place in a college-hosted Hackathon focused on Generative AI. (Aug 2024)
- **National Hackathon on AR/VR Development** – Ranked in the top 10 at a national-level hackathon focused on AR/VR. (Oct 2023)

## Certifications

- **AWS Academy Graduate - AWS Academy Cloud Architecting** – AWS Academy (Sep 2024)
- **AICTE Virtual Internship** – Nalla Narasimha Reddy Education Society's Group of Institutions (2024)
- **Python For Data Science** – IIT Madras (NPTEL) (Jan 2025)
- **The Joy of Computing Using Python** – IIT Madras (NPTEL) (Jul 2023)
- **Supervised Machine Learning: Regression and Classification** – Stanford University (Coursera) (Jun 2023)